

Lab 3: Architectural Pattern Selection – Written Justification

SRN: PES1UG24CS195 | Project: Digital Campus Library Reservation Gateway

Architecture Selection: We chose Layered Architecture for the Digital Campus Library Reservation Gateway.

1. Architectural Choice

Layered Architecture, deployed in the usual web client–server way (browser client, one application server). The server is split into three layers. The **Presentation Layer** holds the Web UI. The **Business Layer** holds Auth Manager (UC-00), Catalog Search (UC-01), Hold Manager (UC-02, 03, 06), Fine Manager (UC-05, 07, 08) and Notification Manager (UC-04). The **Data Layer** holds the Library Database. The Payment Gateway and Notification Service from LAB1 are external components reached only through the IPayment and IMessaging interfaces.

2. Two Reasons for the Choice

- **The hold queue and fines need strict data consistency.** FR-001 requires a FIFO waitlist with correct queue positions, FR-005 blocks holds once fines exceed ₹500, and a member may keep at most 3 active holds. These rules must be checked and updated in one transaction against a single database. Microservices would split this data across services and bring distributed transactions or eventual consistency, which can give two members the same queue position or let a blocked member place a hold. Layered design keeps it simple and correct.
- **The system is campus-scale and its outside dependencies must be replaceable.** LAB1 defines a fixed set of 9 use cases for one institution and a small team, so the operational cost of microservices is not justified. Payment and messaging are wrapped in components behind interfaces, so the college can change its payment provider or SMS/email vendor without touching Hold Manager or Fine Manager. Client–server alone would not structure the server-side logic and would leave one server as a single point of failure (see Performance).

3. Security Advantage

FR-004 requires authentication before any restricted feature. In a layered design the Web UI can reach the system only through the Business Layer interfaces and never touches the database. Restricted operations require a valid session issued by Auth Manager, which also applies role rules (for example, only a Head Librarian may apply a fine waiver, UC-07) and the lockout after 5 failed logins within 15 minutes. Only the Business Layer can reach the Library Database, and card details are handled entirely by the external Payment Gateway, so the system never stores card data.

4. Performance Benefit

NFR-001 needs catalog search on 50,000 titles in under 150 ms. Catalog Search is a separate component with its own ICatalogData access, so it can use indexed, paginated queries and result caching without being slowed by hold or fine logic. For NFR-002 (99.5% uptime), the application tier is stateless, so several identical server instances can run behind a load balancer during peak periods such as semester start. Notification Manager sends messages outside the hold request, so the 5-minute notification target (FR-002) never delays a student's response.

Trade-off Acknowledged

Layers add some call overhead and scale as a whole rather than per feature. For a campus library with tightly linked data and a small team, that cost is small compared with the consistency, security and simplicity gained.